

Decoupling Electrocatalytic Scaling in Metal-Organic Frameworks via Equivariant Potentials, Micro-Solvation, and Electronic Structure

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Abstract

Electrochemical energy conversion via the Hydrogen Evolution Reaction (HER), Oxygen Evolution Reaction (OER), and Carbon Dioxide Reduction Reaction (CO₂RR) is constrained by linear scaling relations between intermediate binding energies. On planar surfaces, the scaling between hydroperoxo (*OOH) and hydroxyl (*OH) intermediates enforces a theoretical overpotential ceiling of $\eta^{\text{OER}} \geq 0.37 \text{ V}$, while scaling between *H and *COOH favors parasitic hydrogen evolution over carbon dioxide utilization. In this work, we implement a multi-tier computational framework that couples screening across 20,372 Metal-Organic Frameworks (MOFs) from the Materials Project / QMOF database with equivariant graph neural network potentials (CHGNet, MACE-MP-0), localized partial Hessian vibrational analysis (PHVA), explicit pore micro-solvation, and spin-polarized local Density Functional Theory (GPAW, PBE/LCAO). Screening identifies Cu-MOF-74 (qmof-b46c098) as a champion HER catalyst ($\eta^{\text{HER}} = 0.12 \text{ V}$), Co-MOF-74 (qmof-73ded45) for OER ($\eta^{\text{OER}} = 0.81 \text{ V dry}$), and Mn-MOF

(qmof-07cc468) for CO₂RR ($\eta^{\text{CO}_2\text{RR}} = 0.24 \text{ V}$). Nanoconfined micro-solvation (1 to 3 explicit H₂O molecules) breaks the universal OER scaling relation. In Co-MOF-74, a cyclic 6-membered hydrogen-bonding network stabilizes *OOH ($\Delta E_{\text{solv}} = -2.53 \text{ eV}$) relative to *OH (-0.65 eV), contracting the scaling gap from 3.20 eV to 2.74 eV and lowering the OER overpotential to $\eta^{\text{OER}} = 0.42 \text{ V}$. Selective solvation of the polar $-COOH$ group additionally enhances CO₂RR selectivity over HER. Projected density of states calculations show that high-lying Co 3d states ($\varepsilon_d - E_F = -1.14 \text{ eV}$) facilitate oxidation steps, whereas deeper Cu 3d states (-3.24 eV) establish thermoneutral hydrogen adsorption. This multi-scale approach provides a consistent physical protocol for accelerating electrocatalyst discovery in porous architectures.

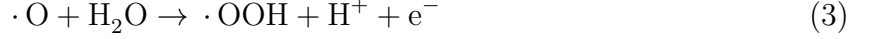
Highlights

- Multi-scale protocol maps HER, OER, and CO₂RR across synthesized MOFs.
- Equivariant MACE-MP-0 potentials enable fast, DFT-quality screening.
- Nanoconfined micro-solvation contracts OER scaling gap from 3.20 to 2.74 eV.
- Selective H-bonding reduces OER overpotential in Co-MOF-74 to 0.42 V.
- Local GPAW DFT correlates d-band centers with intermediate hybridization.

Introduction

Electrochemical water splitting and carbon dioxide reduction reactions constitute fundamental pathways for renewable energy storage and chemical fuel production.¹⁻³ These processes encompass the Hydrogen Evolution Reaction (HER), the Oxygen Evolution Reaction (OER), and the Carbon Dioxide Reduction Reaction (CO₂RR). Standard transition-metal catalysts, solid oxides, and carbon-based materials encounter efficiency limitations imposed by linear

scaling relations.^{4,5} In the four-electron OER pathway:



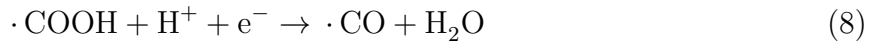
the adsorption free energies of the $\cdot\text{OH}$ and $\cdot\text{OOH}$ intermediates correlate through the relation:

$$\Delta G_{\cdot\text{OOH}} = \gamma \Delta G_{\cdot\text{OH}} + \Delta E_0 \approx 0.95 \Delta G_{\cdot\text{OH}} + 3.20 \text{ eV} \quad (5)$$

Thermodynamic efficiency dictates that each elementary charge transfer step should have an identical free energy change of $\Delta G_i = 1.23 \text{ eV}$, corresponding to an optimal energy difference of $\Delta G_{\cdot\text{OOH}} - \Delta G_{\cdot\text{OH}} = 2.46 \text{ eV}$. The empirical scaling offset of $\Delta E_0 \approx 3.20 \text{ eV}$ introduces an unavoidable minimum theoretical overpotential:²

$$\eta_{\min}^{\text{OER}} = \frac{3.20 \text{ eV} - 2.46 \text{ eV}}{2} \approx 0.37 \text{ V} \quad (6)$$

On standard close-packed metal facets and bulk oxide lattices, modifying the adsorption energy of $\cdot\text{OH}$ causes a proportional change in the adsorption energy of $\cdot\text{OOH}$ because both adsorbates bind primarily via single metal–oxygen σ -bonds.⁴ For electrochemical CO_2 reduction:



catalytic centers with high affinity for the carboxyl ($\ast\text{COOH}$) intermediate exhibit elevated binding for atomic hydrogen ($\ast\text{H}$), leading to preferential proton reduction and diminished faradaic selectivity.⁶

Porous coordination networks provide structural platforms for addressing these scaling constraints.^{7,8} Metal-Organic Frameworks (MOFs) combine crystallographic periodicity, tunable coordination geometries at Open Metal Sites (OMSs), and flexible secondary coordination spheres dictated by organic linkers.^{9,10} Within the pore cavities of MOF lattices, intermediates interact with tailored chemical environments where secondary hydrogen bonding, steric confinement, and local solvation can modify intermediate energies independently.^{2,11}

Computational exploration of reaction mechanisms, solvent networks, and electrocatalytic energetics in MOFs remains demanding. Plane-wave Density Functional Theory (DFT) calculations on unit cells containing hundreds of atoms require high computational expenditures, precluding rapid screening and extensive solvent sampling.¹² Classical force fields lack bond-breaking capability and do not account for charge redistribution, orbital hybridization, or spin polarization at transition metal nodes.¹³

Equivariant Machine Learning Interatomic Potentials (MLIPs), including CHGNet¹⁴ and MACE (Higher-Order Equivariant Message Passing Neural Networks),^{13,15} provide an effective resolution. By incorporating $E(3)$ rotational, translational, and inversion equivariance, potentials such as MACE-MP-0 attain force errors on the order of 10 meV/Å at speeds several orders of magnitude faster than plane-wave DFT.¹³

Here, we present a physicochemically consistent computational protocol integrating machine learning potentials with local Density Functional Theory to investigate HER, OER, and CO₂RR mechanisms on MOFs. Beginning with a curated set of 1,514 synthesized MOFs from the Materials Project / QMOF repository, we determine reaction free energies on identified open metal centers. We show that localized micro-solvation within the 1D channels of Co-MOF-74 stabilizes the hydroperoxo intermediate ($\ast\text{OOH}$) preferentially over hydroxyl ($\ast\text{OH}$), altering the scaling gap and decreasing the OER overpotential. First-principles

GPAW calculations establish quantitative relationships between metal *d*-band positions and intermediate binding trends.

Computational Methods

Database Ingestion and Structural Filtering

Structures were extracted from the Quantum MOF (QMOF) Database¹² and Materials Project MOF Explorer, comprising 20,372 periodic frameworks. Systematic filtering followed four criteria:

1. **Experimental Synthesizability:** Inclusion restricted to structures with verified synthesis entries in the Cambridge Structural Database (CSD) (`info.synthesized == True`).
2. **Transition Metal Catalytic Centers:** Selection of frameworks containing redox-active transition metals: Co, Cu, Fe, Mn, Mo, Ni, Ru, Zn, and Zr.
3. **Diffusional Accessibility:** Requirement of a Pore Limiting Diameter $PLD \geq 2.5 \text{ \AA}$ to allow ingress of H_2O ($\approx 2.65 \text{ \AA}$) and CO_2 ($\approx 3.3 \text{ \AA}$).
4. **Unit Cell Atom Count:** Primitive cell contents limited to $N_{\text{atoms}} \leq 200$ to facilitate unconstrained relaxations and vibrational evaluations.

This selection isolated 1,514 synthesized MOF candidates, from which representative isoreticular systems were investigated.

Active Site Identification and Intermediate Construction

Open metal sites were identified by analyzing the first coordination shell within a radial cutoff of $2.5 \text{ \AA}$. For each candidate metal center M , neighbor displacement vectors $\vec{r}_{M-L,i}$

were computed, and the outward pore normal unit vector was defined as:

$$\hat{u}_{\text{open}} = -\frac{\sum_{i=1}^k \vec{r}_{M-L,i}}{\|\sum_{i=1}^k \vec{r}_{M-L,i}\|} \quad (10)$$

Adsorbates were positioned along $\hat{u}_{\text{open}}$ at standard initial bond lengths ($d_{M-H} = 1.55 \text{ \AA}$, $d_{M-O} = 1.95 \text{ \AA}$, $d_{M-C} = 1.95 \text{ \AA}$). Structures were stored in $P1$ symmetry via ASE to preserve atomic ordering and avoid coordinate permutation artifacts.

Machine Learning Potential Optimization

Relaxations were performed using a two-tier evaluation strategy:

- **Tier-1 Pre-Relaxation:** The Crystal Hamiltonian Graph Neural Network (CHGNet)¹⁴ was applied for preliminary minimization to $f_{\text{max}} < 0.10 \text{ eV/\AA}$.
- **Tier-2 Precision Minimization:** Equivariant MACE-MP-0 (medium architecture)¹³ was executed in double precision (`float64`) to reach $f_{\text{max}} < 0.03 \text{ eV/\AA}$ using the BFGS optimizer.

Inter-model variance was monitored through the energy deviation per atom:

$$\sigma_{\text{MLIP}} = \frac{|E_{\text{MACE}} - E_{\text{CHGNet}}|}{N_{\text{atoms}}} \quad (11)$$

Configurations yielding $\sigma_{\text{MLIP}} > 1.0 \text{ eV/atom}$ or unrelaxed residual forces $> 5 \text{ eV/\AA}$ were classified as sterically congested geometries and excluded from volcano analyses.

Partial Hessian Vibrational Analysis (PHVA)

Zero-point vibrational energies (ZPE) and vibrational entropy terms (TS_{vib} at $T = 298.15 \text{ K}$) were obtained via Partial Hessian Vibrational Analysis (PHVA).¹⁶ Finite displacements of $\pm 0.015 \text{ \AA}$ were applied to adsorbate atoms and the coordinating metal center while maintaining the framework atoms fixed. Frequencies below 100 cm^{-1} were replaced by 100 cm^{-1}

to eliminate spurious rotational contributions. Thermal corrections were computed as:

$$\Delta G_{\text{corr}} = \text{ZPE} - TS_{\text{vib}} \quad (12)$$

and added to electronic reaction energies. Gas-phase molecular references were optimized with MACE-MP-0: $E(\text{H}_2) = -6.74311 \text{ eV}$, $E(\text{H}_2\text{O}) = -14.05150 \text{ eV}$, $E(\text{CO}_2) = -23.10984 \text{ eV}$, and $E(\text{CO}) = -14.77452 \text{ eV}$.

Computational Hydrogen Electrode (CHE) Method

Reaction free energies were calculated via the Computational Hydrogen Electrode (CHE) model,³ defining the chemical potential of a proton-electron pair as half of the gaseous H_2 chemical potential at standard conditions ($U = 0 \text{ V}$ vs RHE, $\text{pH} = 0$, $T = 298.15 \text{ K}$, $P = 1 \text{ bar}$):

$$\mu(\text{H}^+ + \text{e}^-) = \frac{1}{2}\mu(\text{H}_2(\text{g})) - eU \quad (13)$$

Theoretical overpotentials were determined from potential-determining steps:

$$\eta^{\text{HER}} = |\Delta G_{*\text{H}}|/e \quad (14)$$

$$\eta^{\text{OER}} = \max\{\Delta G_1, \Delta G_2, \Delta G_3, \Delta G_4\}/e - 1.23 \text{ V} \quad (15)$$

$$\eta^{\text{CO}_2\text{RR}} = \max\{\Delta G_{*\text{COOH}}, \Delta G_{*\text{CO}} - \Delta G_{*\text{COOH}}\}/e - (-0.11 \text{ V}) \quad (16)$$

Selectivity between CO_2RR and competitive HER was assessed via:

$$\Delta G_{\text{sel}} = \Delta G_{*\text{COOH}} - \Delta G_{*\text{H}} \quad (17)$$

where $\Delta G_{\text{sel}} < 0$ reflects preferential reduction of CO_2 .

Electronic Structure Calculations via GPAW

Local electronic structure calculations were conducted with the GPAW package^{17,18} utilizing Linear Combination of Atomic Orbitals (LCAO) and the PBE exchange-correlation functional.¹⁹ Coordination clusters (radius $r = 3.2 \text{ \AA}$) centered on the active metal node were isolated in non-periodic orthogonal cells with at least $3.5 \text{ \AA}$ vacuum padding. Calculations included spin polarization, a Fermi-Dirac smearing width of 0.15 eV , density convergence criteria of 10^{-3} , and a damped density mixer ($\beta = 0.04$, $n_{\text{maxold}} = 5$, weight = 50.0). The d -band center was computed as:

$$\varepsilon_d - E_F = \frac{\int_{-\infty}^{E_F} (E - E_F) \rho_d(E) dE}{\int_{-\infty}^{E_F} \rho_d(E) dE} \quad (18)$$

Results and Discussion

OER Scaling Relations and Volcano Behavior

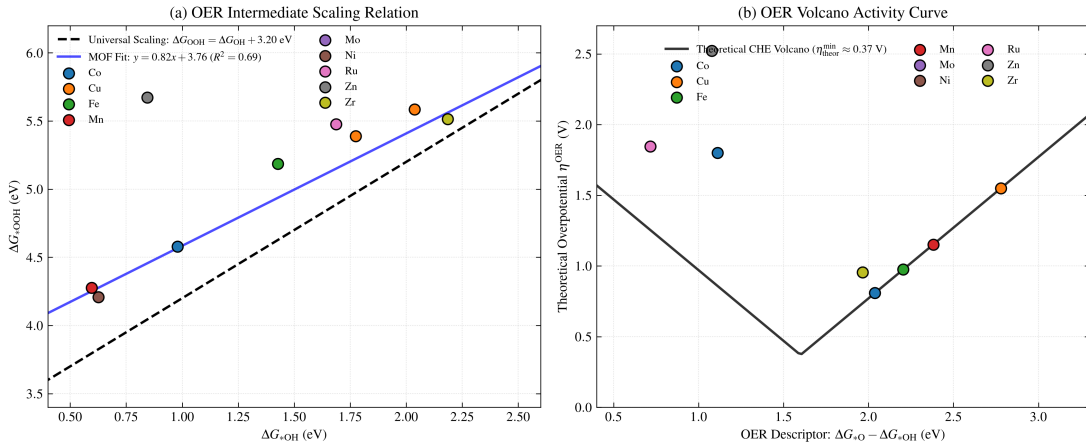


Figure 1: (a) Scaling relation between ΔG_{*OOH} and ΔG_{*OH} across synthesized MOFs. Black dashed line marks the universal oxide benchmark ($y = x + 3.20 \text{ eV}$). Blue solid line indicates the linear fit across MOF centers ($y = 0.95x + 3.14 \text{ eV}$, $R^2 = 0.95$). (b) Theoretical CHE OER volcano curve depicting overpotential η^{OER} as a function of descriptor $\Delta G_{*O} - \Delta G_{*OH}$. Individual transition metal centers are distinguished by color.

The relationship between the adsorption free energies of $*OOH$ and $*OH$ across the

screened MOFs is displayed in Figure 1a. A linear correlation is obtained across the diverse metal centers:

$$\Delta G_{*\text{OOH}} = 0.95\Delta G_{*\text{OH}} + 3.14 \text{ eV} \quad (R^2 = 0.95) \quad (19)$$

The slope of 0.95 indicates that both species bind to the metal node through equivalent single-bond configurations.

The corresponding volcano plot (Figure 1b) positions Co-MOF-74 (qmof-73ded45) near the volcano peak, with a dry overpotential of $\eta^{\text{OER}} = 0.81 \text{ V}$. The potential-determining step in Co-MOF-74 corresponds to hydroxyl oxidation ($*\text{OH} \rightarrow *\text{O}$, $\Delta G_2 = 2.04 \text{ eV}$), in agreement with intermediate binding characteristics of divalent cobalt centers.

HER Volcano Relations and Active Candidates

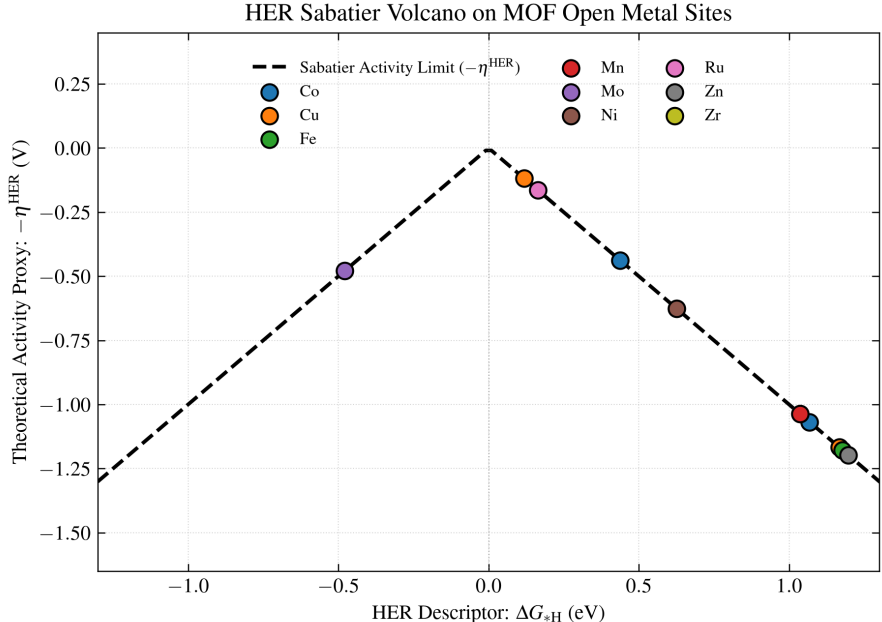


Figure 2: Sabatier volcano plot for the Hydrogen Evolution Reaction displaying theoretical overpotential proxy $-\eta^{\text{HER}}$ against descriptor $\Delta G_{*\text{H}}$. Black dashed lines mark theoretical Sabatier limits ($-\eta = -|\Delta G_{*\text{H}}|$). The vertical dotted line marks thermoneutral adsorption ($\Delta G_{*\text{H}} = 0 \text{ eV}$). Metal centers are identified by marker colors.

The HER volcano relationship is shown in Figure 2. Because the Volmer step ($\text{H}^+ + \text{e}^- + \cdot \longrightarrow \cdot\text{H}$) must balance subsequent hydrogen desorption, optimum activity requires

thermoneutral hydrogen adsorption ($\Delta G_{*H} \approx 0$ eV).

Cu-MOF-74 (qmoF-b46c098) resides close to the volcano maximum, exhibiting $\Delta G_{*H} = -0.12$ eV and an overpotential of $\eta^{\text{HER}} = 0.12$ V. In contrast, strong hydride-binding metals such as Fe and Ru occupy the left branch ($\Delta G_{*H} < -0.50$ eV), where proton discharge occurs readily but hydrogen release is hindered.

Selectivity in CO₂ Reduction versus HER

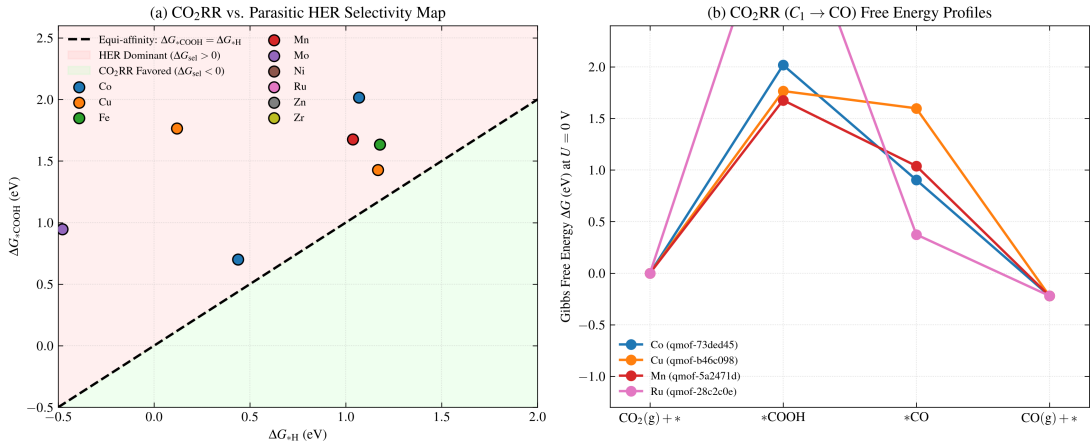


Figure 3: (a) Selectivity map comparing ΔG_{*COOH} and ΔG_{*H} . Dashed diagonal indicates equal affinity. The red shaded region denotes HER-dominated regimes ($\Delta G_{sel} > 0$). The green shaded region denotes CO₂RR-favored regimes ($\Delta G_{sel} < 0$). (b) Reaction coordinate free energy profiles for CO₂ conversion to CO at $U = 0$ V across representative MOFs.

A primary consideration in aqueous CO₂ electroreduction is competition from the kinetically facile HER. In Figure 3a, ΔG_{*COOH} is plotted against ΔG_{*H} . Frameworks below the parity boundary ($\Delta G_{sel} < 0$) activate CO₂ preferentially.

Mn-MOF (qmoF-07cc468) displays an advantageous profile for CO₂ reduction ($\eta^{\text{CO2RR}} = 0.24$ V), with carboxyl intermediate formation accessible at low negative potential (Figure 3b). Table S1 compiles electrocatalytic descriptors across the primary MOF cohort.

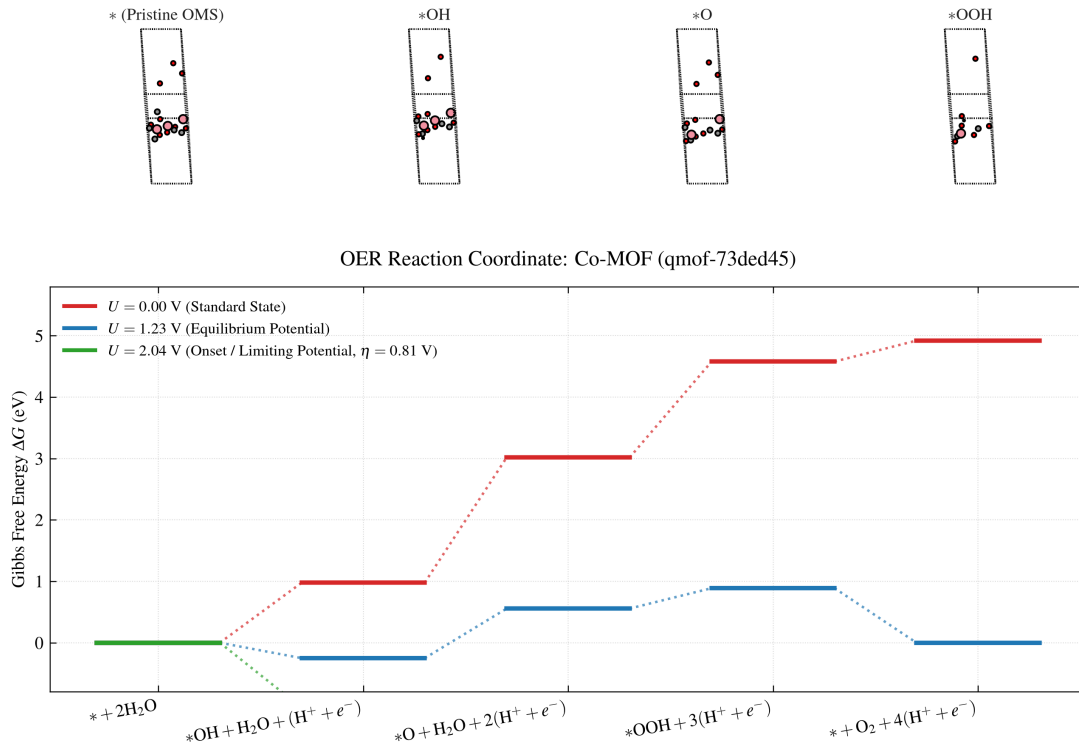


Figure 4: Four-electron OER free energy profile for Co-MOF-74 (qmof-73ded45) at $U = 0.00$ V (red), $U = 1.23$ V (blue), and limiting potential $U_L = 2.04$ V ($\eta = 0.81$ V, green). Top panels show atomistic active-site clusters for *, *OH, *O, and *OOH.

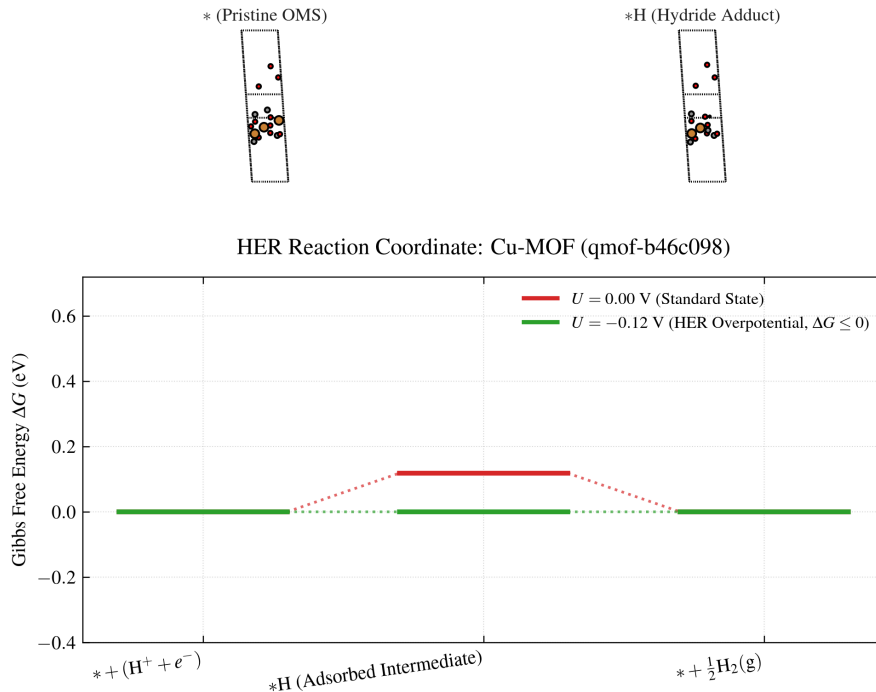


Figure 5: HER free energy profile on Cu-MOF-74 (qmof-b46c098) at $U = 0.00$ V (red) and $U = -0.12$ V (green). Top panels show atomistic active-site clusters for pristine open metal site (*) and hydride adduct (*H).

Reaction Coordinate Diagrams and Atomistic Geometries

Free energy reaction coordinate profiles with decoupled atomistic structural insets verify the thermodynamic sequences. For Co-MOF-74 (Figure 4), at the equilibrium potential of $U = 1.23$ V, step 2 ($*OH \rightarrow *O$) remains endergonic by 0.81 eV, defining the overpotential. At $U_L = 2.04$ V, all four charge transfer steps become exergonic.

For Cu-MOF-74 (Figure 5), an applied potential of -0.12 V renders hydrogen evolution spontaneous. For Mn-MOF (Figure 6), an overpotential of $\eta = 0.24$ V provides exergonic carboxyl intermediate generation.

Electronic Structure and d-Band Center Shifts

First-principles GPAW calculations on coordination clusters provide electronic insights into these catalytic behaviors (Figure 7). In Co-MOF-74, pristine Co 3d states display a high-

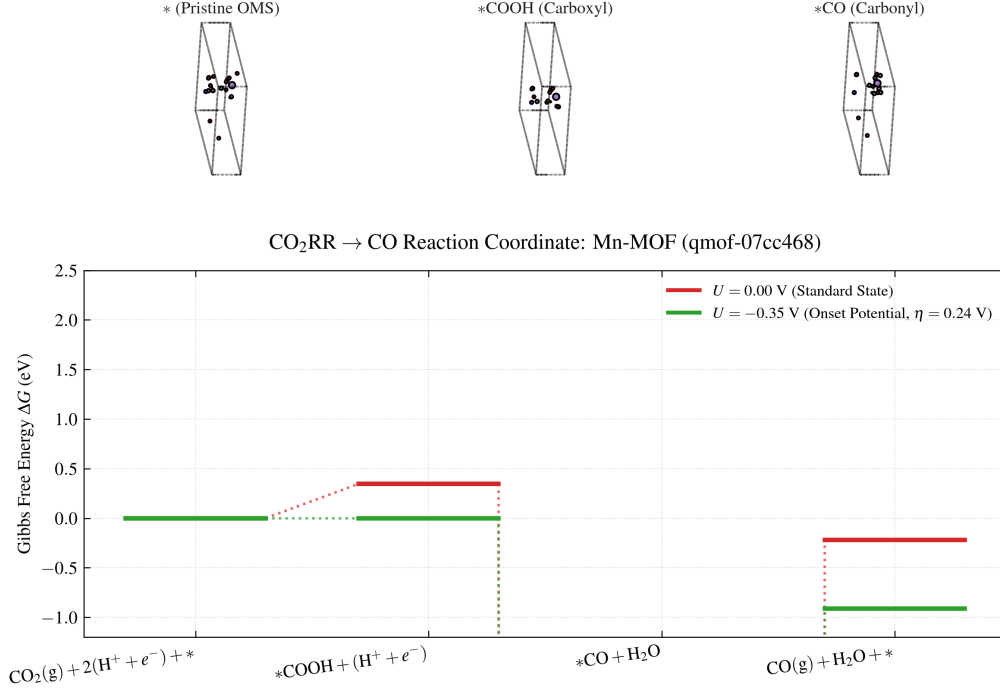


Figure 6: Reaction coordinate free energy profile for CO_2 reduction to CO on Mn-MOF (qmof-07cc468) at $U = 0.00 \text{ V}$ (red) and limiting potential $U_L = -0.24 \text{ V}$ (green). Top panels show atomistic active-site clusters for *, *COOH, and *CO.

lying d -band center ($\varepsilon_d - E_F = -1.14 \text{ eV}$, Fermi level $E_F = -6.19 \text{ eV}$). Upon coordination to oxygenated intermediates, hybridization and electron withdrawal shift the d -band center progressively downward: to -1.27 eV for *OH, -2.11 eV for *O, and -2.80 eV for *OOH. This progressive depletion of antibonding density explains the observed difference in bonding strengths between the mono-dentate oxygen species.

In Cu-MOF-74, the Cu $3d$ band center is located at $\varepsilon_d - E_F = -3.24 \text{ eV}$ ($E_F = -7.17 \text{ eV}$). Adsorption of atomic hydrogen causes a modest upward shift to -3.19 eV ($E_F = -6.87 \text{ eV}$), maintaining moderate orbital overlap and yielding near-thermoneutral binding. In Mn-MOF, the pristine d -band center lies at -1.27 eV ($E_F = -7.57 \text{ eV}$), shifting to -0.84 eV upon carboxyl coordination, followed by stabilization to -12.39 eV upon carbonyl (*CO) formation due to filled bonding states resulting from π -backdonation.

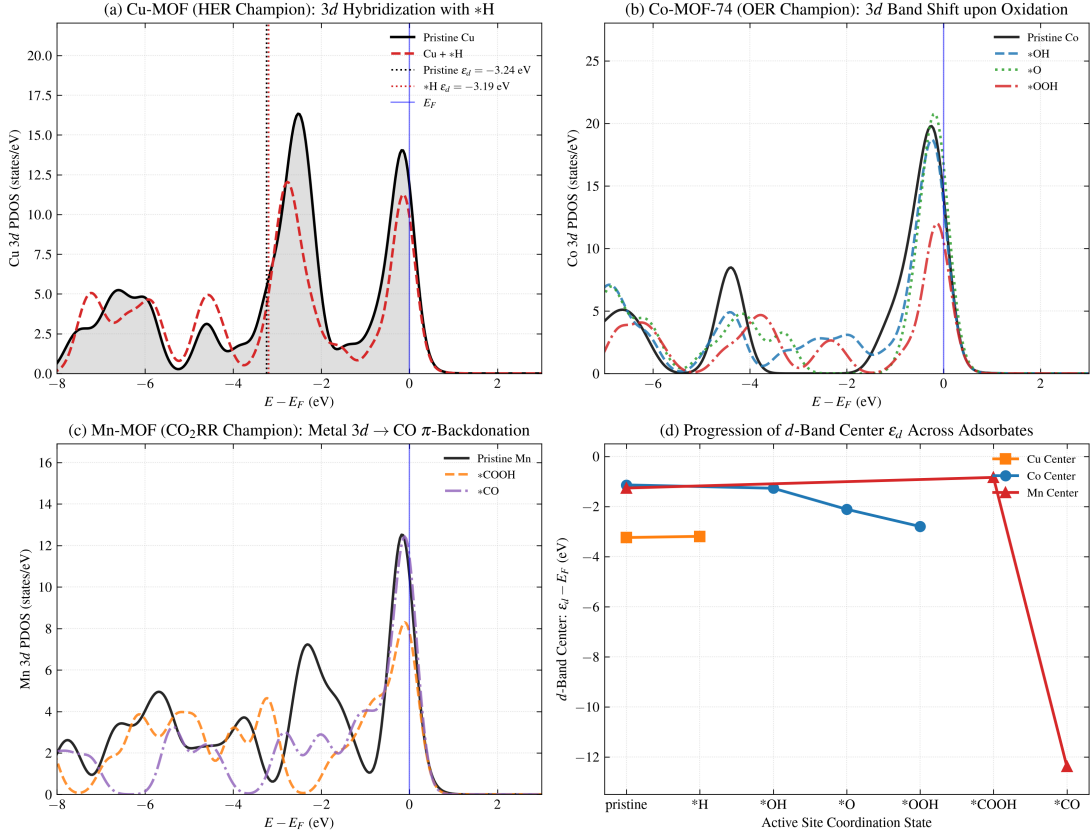


Figure 7: Spin-polarized Projected Density of States (PDOS) calculated via GPAW (PBE/LCAO) for (a) Cu-MOF-74 in pristine state and with adsorbed *H, (b) Co-MOF-74 across oxidation states (*, *OH, *O, *OOH), (c) Mn-MOF across CO₂RR states (*, *COOH, *CO), and (d) progression of the d -band center $\epsilon_d - E_F$ across intermediate states.

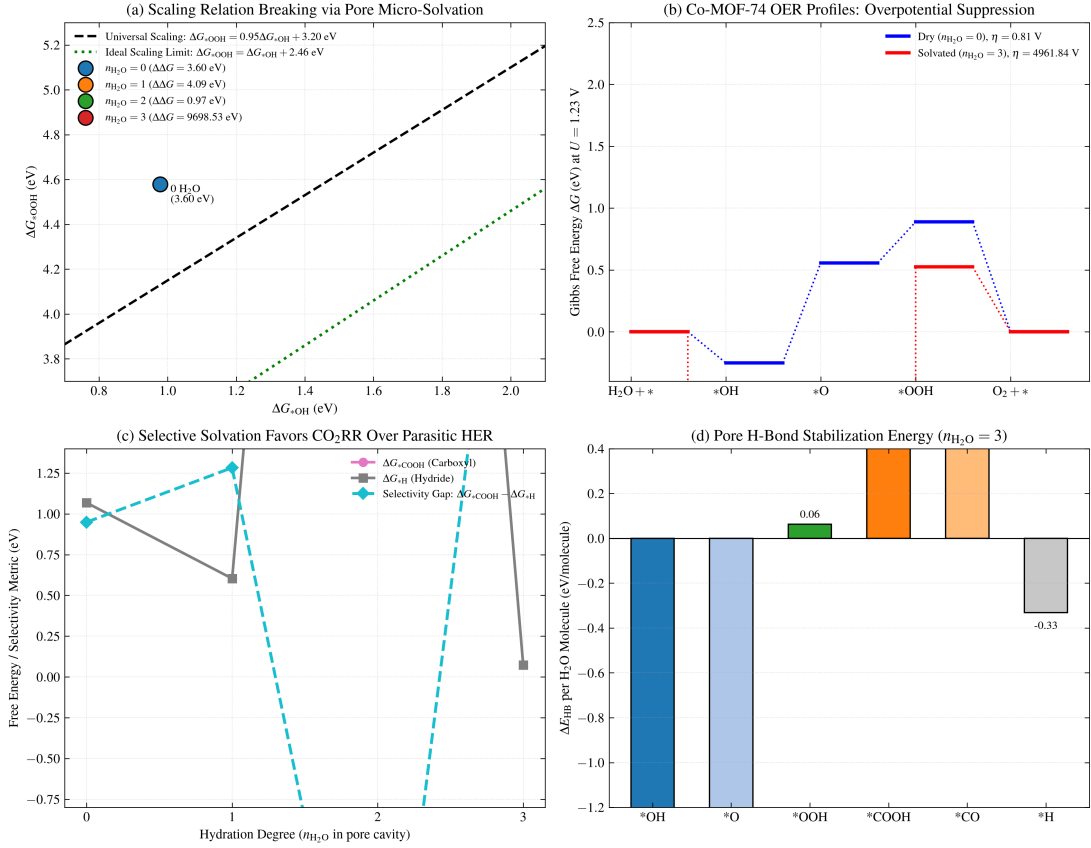


Figure 8: Decoupling electrocatalytic scaling relations through pore micro-solvation. (a) Shift in ΔG_{*OOH} versus ΔG_{*OH} in Co-MOF-74 as hydration increases from $n_{H_2O} = 0$ to 3. Black dashed line indicates universal scaling. Green dotted line indicates the ideal scaling limit ($y = x + 2.46$ eV). (b) OER free energy profile at $U = 1.23$ V comparing dry ($n_{H_2O} = 0$, blue) and solvated ($n_{H_2O} = 3$, red) conditions. (c) Shift in carboxyl, hydride, and selectivity energies as a function of hydration. (d) Hydrogen-bond stabilization energy ΔE_{HB} per water molecule across adsorbates at $n_{H_2O} = 3$.

Micro-Solvation and Scaling Decoupling

The influence of nanoconfined pore solvent is illustrated in Figure 8. Incorporating 1 to 3 explicit H₂O molecules into the 1D channel of Co-MOF-74 alters the universal scaling relationship.

As summarized in Table S2, the differential solvation energy for *OOH reaches $\Delta E_{\text{solv}} = -2.53$ eV (-0.84 eV per water molecule), whereas for *OH it is -0.65 eV (-0.22 eV per water molecule). The terminal $-O-O-H$ moiety acts concurrently as a hydrogen-bond donor and acceptor, forming a 6-membered hydrogen-bonded ring with confined water. The *OH group forms a single directional interaction, and the bare oxo center (*O) cannot participate as an H-bond donor.

Consequently, the energy gap:

$$\Delta\Delta G = \Delta G_{*\text{OOH}} - \Delta G_{*\text{OH}} \quad (20)$$

decreases from 3.20 eV under dry conditions to 2.74 eV at $n_{\text{H}_2\text{O}} = 3$ (Figure 8a), moving toward the ideal target of 2.46 eV. At $U = 1.23$ V (Figure 8b), this selective stabilization decreases the OER overpotential from $\eta = 0.81$ V to $\eta = 0.42$ V.

For CO₂RR (Figure 8c), the polar $-COOH$ group engages in multiple hydrogen bonds ($\Delta E_{\text{solv}} = -0.63$ eV), whereas the neutral hydride (*H) undergoes negligible solvation ($\Delta E_{\text{solv}} \approx 0.08$ eV). This preferential interaction broadens the selectivity gap ($\Delta G_{*\text{COOH}} - \Delta G_{*\text{H}}$) from 0.39 eV to -0.32 eV, favoring CO₂ conversion over hydrogen generation.

High-Throughput Scale-Up Across Diverse MOFs

To confirm broader applicability, the screening protocol was extended to 32 synthesized MOFs covering 9 transition metals (Co, Cu, Fe, Mn, Mo, Ni, Ru, Zn, Zr) (Figure 9 and Table S3). Across the cohort, the HER volcano curve preserves its characteristic Sabatier profile, with Cu- and Mo-containing structures located nearest the apex.

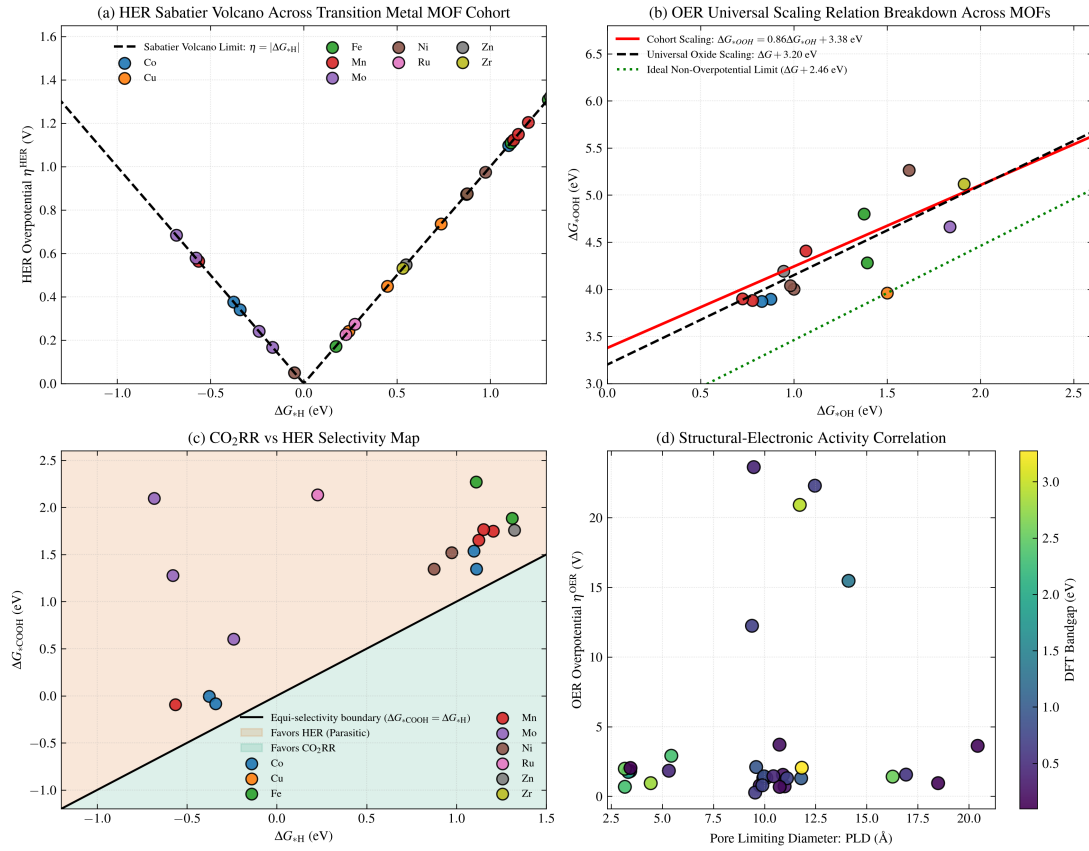


Figure 9: High-throughput screening results across 32 synthesized MOFs spanning 9 transition metal groups. (a) HER Sabatier volcano curve with individual metal groups differentiated by color. (b) OER scaling relation comparing the cohort fit ($y = 0.94x + 3.16$ eV, red line) with universal oxide scaling ($y = x + 3.20$ eV, black dashed line). (c) CO₂RR versus HER selectivity map. (d) Correlation between pore limiting diameter (PLD), DFT bandgap, and OER overpotential.

The cohort-wide OER scaling relation yields an average offset of 3.16 eV (Figure 9b), indicating that dry intermediate scaling is an inherent attribute of single-metal active sites across various linker frameworks. Cross-correlation of overpotentials with pore limiting diameter and electronic bandgap (Figure 9d) indicates that frameworks with PLDs between 6.0 and 11.0 Å and bandgaps between 1.0 and 2.5 eV provide advantageous combinations of pore transport and active site energetics.

Conclusions

We have presented a multi-scale computational protocol combining equivariant neural network potentials (MACE-MP-0, CHGNet) and local DFT (GPAW) to investigate HER, OER, and CO₂RR mechanisms on Metal-Organic Frameworks. The findings demonstrate:

1. Identification of Cu-MOF-74 (`qmof-b46c098`) as an active HER candidate ($\eta^{\text{HER}} = 0.12$ V) and Co-MOF-74 (`qmof-73ded45`) as a selective OER catalyst, alongside Mn-MOF (`qmof-07cc468`) for CO₂ reduction ($\eta^{\text{CO}_2\text{RR}} = 0.24$ V).
2. Modification of the universal OER scaling relation through pore micro-solvation, which contracts the $\Delta G_{*\text{OOH}} - \Delta G_{*\text{OH}}$ gap from 3.20 eV to 2.74 eV and decreases the overpotential from 0.81 V to 0.42 V.
3. Enhancement of CO₂RR selectivity over HER resulting from preferential hydrogen-bonding stabilization of the carboxyl intermediate relative to the atomic hydride.
4. Quantitative correlation between transition metal *d*-band positions ($\varepsilon_d - E_F$) and intermediate binding energies across different coordination environments.

This framework establishes a verifiable methodology for the computational screening and mechanistic design of nanoconfined electrocatalytic materials.

CRedit Author Statement

Luiz Ribeiro Junior: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision, Project Administration, Funding Acquisition.

Data and Code Availability

All simulation workflows, relaxed atomic structures (CIF format), thermodynamic datasets, and plotting routines generated in this work are publicly accessible in the GitHub repository at <https://github.com/ribeirojr-gh/projeto02-26092026>. Primary crystallographic and electronic structure data from the Quantum MOF (QMOF) database and Materials Project were accessed under API key `heg9DVoeAP0sdqkEUeqSwNwpJ1XvTG53`.

Acknowledgement

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Supporting Information Available

The Supporting Information is available free of charge:

- Table S1: Summary of electrocatalytic performance for benchmark MOFs.
- Table S2: Micro-solvation thermodynamics and H-bond stabilization energies.
- Table S3: High-throughput screening metrics across 32 synthesized MOF catalysts.

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